Homework 2: Rainfall in southern Italy

Spatial statistics 2026–2027 · Professor: Dr. Lasinio

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This report studies annual rainfall in 1970. I first check the observations, station locations and environmental relationships. Then I choose the IDW power by cross-validation and predict rainfall at the supplied land-grid locations. The code and explanations are together, so the results can be checked directly.

How to run this file

Keep this file beside the `data` folder containing `RainData.RData` and `AllGrids.RData`. Open it in RStudio and click **Knit to PDF**. The plots appear inside the report. The analysis does not write separate plot files or CSV files; the final predictions remain in the R object `prediction_grid` when the chunks are run interactively. Knitting creates the report and its temporary supporting files. HTML output is also available from the Knit menu.

Install the packages below once, if needed. PDF compilation also needs a LaTeX installation; `tinytex::install_tinytex()` is one option.

```
install.packages(c("gstat", "sp", "knitr", "rmarkdown", "tinytex"))
# Only if a LaTeX installation is not already available:
tinytex::install_tinytex()
```

1 Data checks and choice of year

1.1 What is actually in the two files?

I load each file into its own environment. This avoids replacing my objects or random-number seed with objects stored in an R workspace.

```
needed <- c("gstat", "sp", "knitr")
stopifnot(all(vapply(needed, requireNamespace, logical(1), quietly = TRUE)))
rain_file <- new.env()
grid_file <- new.env()
invisible(load("data/RainData.RData", envir = rain_file))
invisible(load("data/AllGrids.RData", envir = grid_file))
rain_all <- rain_file$dati
land_grid <- rain_file$grid
marine_example <- grid_file$grid_1995

file_check <- data.frame(
  Object = c("RainData: dati", "RainData: grid", "AllGrids: grid_1995"),
  Rows = c(nrow(rain_all), nrow(land_grid), nrow(marine_example)),
  Content = c("Rainfall, 1970-1982", "Land grid with elevation",
              "Marine grid; other years 1996-2010")
)
knitr::kable(file_check)
```

Object	Rows	Content
RainData: dati	2373	Rainfall, 1970–1982
RainData: grid	1005	Land grid with elevation
AllGrids: grid_1995	517	Marine grid; other years 1996–2010

The two uploaded files do not describe the same prediction task. The objects in `AllGrids.RData` contain bathymetry, salinity and sea temperature, with years 1995–2010. Their coordinates are named `X` and `Y`; the rainfall file uses `XUTM` and `YUTM`. I cannot safely combine them just by changing column names or units. Their date ranges also do not overlap.

Fortunately, `RainData.RData` already includes the supplied land grid `grid`, with 1,005 locations and elevation. It is the concatenation of `Basquota32` and `Calabriaquota32`. I use this existing grid, without creating a replacement regular grid. The objects `Bas` and `Cal` provide regional outline coordinates for the maps; `grid.int` is their concatenated outline, not the prediction grid.

```
expected_grid <- rbind(rain_file$Basquota32, rain_file$Calabriaquota32)
stopifnot(isTRUE(all.equal(land_grid, expected_grid,
                           check.attributes = FALSE)))
stopifnot(isTRUE(all.equal(rain_file$grid.int,
  rbind(rain_file$Bas, rain_file$Cal)[c("XUTM", "YUTM")],
  check.attributes = FALSE)))
stopifnot(!anyNA(land_grid),
          !anyDuplicated(land_grid[c("XUTM", "YUTM")]))
```

1.2 Correcting the Puglia label

The course hint identifies one Puglia label that should be Calabria. I check the three Puglia-labelled stations against the supplied Calabria outline. Only station **4367**, at approximately **38.55 N, 16.02 E**, is inside it. I correct that station's label across all years. I keep its coordinates, elevation and rainfall values unchanged. The other two Puglia-labelled stations remain in the dataset, and all stations in the chosen year enter the analysis.

```
station_info <- unique(rain_all[c("ID", "Nome_Regione", "Lat10",
                                  "Lon10", "XUTM", "YUTM")])
puglia <- subset(station_info, Nome_Regione == "Puglia")
puglia$inside_Calabria <- sp::point.in.polygon(
  puglia$XUTM, puglia$YUTM,
  rain_file$Cal$XUTM, rain_file$Cal$YUTM
) > 0
knitr::kable(puglia[c("ID", "Lat10", "Lon10", "inside_Calabria")],
             digits = 3)
```

	ID	Lat10	Lon10	inside_Calabria
5	4085	40.817	16.419	FALSE
6	4086	40.900	16.503	FALSE
150	4367	38.550	16.019	TRUE

```
wrong_id <- puglia$ID[puglia$inside_Calabria]
stopifnot(length(wrong_id) == 1, wrong_id == 4367)
rain_all$Nome_Regione <- as.character(rain_all$Nome_Regione)
fix_rows <- rain_all$ID == wrong_id
rain_all$Nome_Regione[fix_rows] <- "Calabria"
stopifnot(nrow(rain_all) == nrow(rain_file$dati),
          identical(rain_all$totanno, rain_file$dati$totanno))
```

This changes 13 annual labels, all belonging to the same station. The original uploaded file is preserved. I do not automatically reclassify every point near a regional boundary, because a simple outline can be imprecise.

1.3 Why I chose 1970

I choose the year with the largest number of complete station records. If years tie, I take the earliest. This gives a clear rule that does not depend on the rainfall pattern or prediction score. Complete annual records do not prove that every daily measurement used to calculate the annual total was available.

```
fields <- c("ID", "XUTM", "YUTM", "Quota", "totanno")
year_check <- do.call(rbind, lapply(sort(unique(rain_all$anno)),
  function(yr) {
    z <- subset(rain_all, anno == yr)
    data.frame(Year = yr, Stations = nrow(z),
               Complete = sum(complete.cases(z[fields])))
  }))
knitr::kable(year_check, row.names = FALSE)
```

Year	Stations	Complete
1970	189	189
1971	189	189
1972	187	187
1973	188	188
1974	187	187
1975	182	182
1976	181	181
1977	183	183
1978	182	182
1979	181	181
1980	178	178
1981	173	173

Year	Stations	Complete
1982	173	173

```
chosen_year <- year_check$Year[which.max(year_check$Complete)]
dat <- subset(rain_all, anno == chosen_year)
stopifnot(chosen_year == 1970, !anyNA(dat),
          !anyDuplicated(dat$ID),
          !anyDuplicated(dat[c("XUTM", "YUTM")]),
          all(dat$totanno > 0), all(dat$Quota >= 0))
dat$rain <- dat$totanno
dat$x <- dat$XUTM / 1000
dat$y <- dat$YUTM / 1000
prediction_grid <- land_grid
prediction_grid$x <- prediction_grid$XUTM / 1000
prediction_grid$y <- prediction_grid$YUTM / 1000
```

1970 and 1971 tie with 189 complete records. I use **1970**, with **189 stations**. There are no missing values or duplicate locations in this subset. Other years are retained in the source file but are not pooled with 1970: repeated years at the same station are not independent spatial sites.

The response is `totanno`, the annual rainfall total. `Quota` is elevation; region is a geographical grouping, while coordinates describe location. The files contain no formal units or CRS metadata. I interpret rainfall as **mm**, elevation as **m**, and the supplied UTM-style coordinates as **m**, consistent with the variable names and magnitudes. These are working unit assumptions, not metadata supplied by the file. I divide both coordinate axes by 1,000 for distances in km and retain their supplied common coordinate system. I do not assign an unverified EPSG code or calculate Euclidean distances in longitude/latitude. A uniform change from metres to kilometres leaves IDW weights unchanged after normalization.

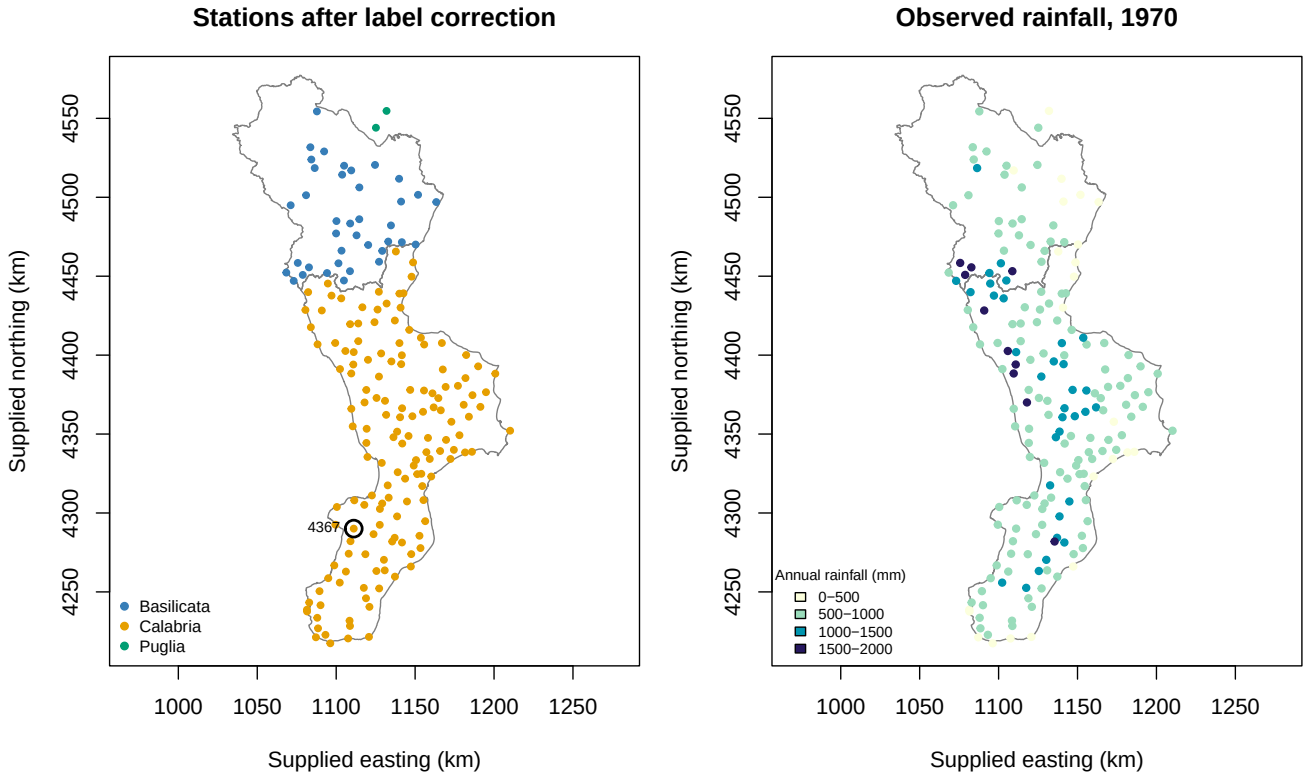
2 Exploratory analysis

2.1 Station map and rainfall distribution

```
region_col <- c(Basilicata = "#377EB8", Calabria = "#E69F00",
                Puglia = "#009E73")
draw_outlines <- function() {
  lines(rain_file$Bas$XUTM / 1000, rain_file$Bas$YUTM / 1000,
        col = "grey50")
  lines(rain_file$Cal$XUTM / 1000, rain_file$Cal$YUTM / 1000,
        col = "grey50")
}
map_frame <- function(title) {
  plot(prediction_grid$x, prediction_grid$y, type = "n", asp = 1,
       xlim = range(c(prediction_grid$x, dat$x)),
       ylim = range(c(prediction_grid$y, dat$y)),
       xlab = "Supplied easting (km)", ylab = "Supplied northing (km)",
       main = title)
  draw_outlines()
}
rain_breaks <- pretty(range(dat$rain), n = 5)
rain_col <- hcl.colors(length(rain_breaks) - 1, "YlGnBu", rev = TRUE)
rain_colour <- function(z) {
  rain_col[cut(z, breaks = rain_breaks, include.lowest = TRUE)]
}
rain_legend <- function() {
  labs <- paste(head(rain_breaks, -1), tail(rain_breaks, -1), sep = "–")
```

```
  legend("bottomleft", legend = labs, fill = rain_col,
         title = "Annual rainfall (mm)", cex = 0.65, bty = "n")
}
```

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
map_frame("Stations after label correction")
points(dat$x, dat$y, pch = 19, cex = 0.65,
       col = region_col[dat$Nome_Regione])
legend("bottomleft", legend = names(region_col), col = region_col,
       pch = 19, bty = "n", cex = 0.75)
i <- which(dat$ID == wrong_id)
points(dat$x[i], dat$y[i], pch = 1, cex = 1.7, lwd = 2)
text(dat$x[i], dat$y[i], "4367", pos = 2, cex = 0.7)
map_frame("Observed rainfall, 1970")
points(dat$x, dat$y, pch = 19, cex = 0.7, col = rain_colour(dat$rain))
rain_legend()
```



The stations cover an elongated area, with more stations in Calabria than in Basilicata. Their spacing is uneven. This matters because IDW uses nearby observations, so the information available for prediction changes across the map. The circled station is the corrected record, not a removed observation. Both outlines are supplied with the rainfall data; they are not a complete administrative map of Italy.

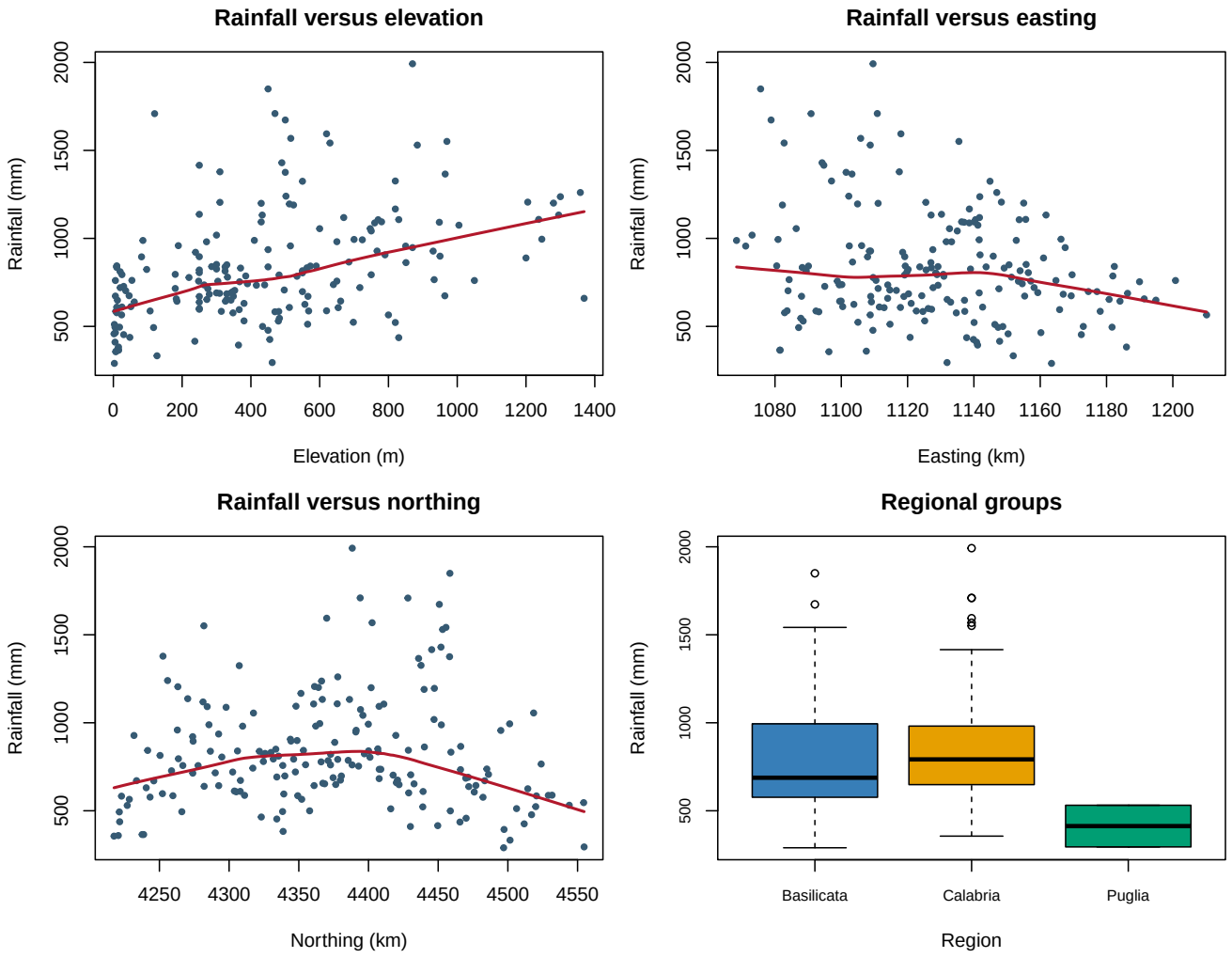
```
rain_summary <- data.frame(
  Minimum = min(dat$rain), Q1 = unname(quantile(dat$rain, 0.25)),
  Median = median(dat$rain), Mean = mean(dat$rain),
  Q3 = unname(quantile(dat$rain, 0.75)), Maximum = max(dat$rain),
  SD = sd(dat$rain)
)
knitr::kable(rain_summary, digits = 1)
```

Minimum	Q1	Median	Mean	Q3	Maximum	SD
289.8	610.2	761.6	829.1	988.3	1992.2	317.3

The mean is **829.1 mm** and the median is **761.6 mm**. The higher mean and the upper tail suggest right-skewness. I keep the large rainfall values: a high measurement alone is not evidence of a recording error.

2.2 Relationships with elevation and location

```
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
for (v in c("Quota", "x", "y")) {
  axis_name <- c(Quota = "Elevation (m)", x = "Easting (km)",
                 y = "Northing (km)")[v]
  plot(dat[[v]], dat$rain, pch = 19, col = "#365A73", cex = 0.65,
       xlab = axis_name, ylab = "Rainfall (mm)",
       main = paste("Rainfall versus", tolower(sub(" \\(.*", "", axis_name))))
  lines(lowess(dat[[v]], dat$rain), col = "#B2182B", lwd = 2)
}
boxplot(rain ~ Nome_Regione, data = dat, col = region_col,
        ylab = "Rainfall (mm)", xlab = "Region", main = "Regional groups",
        cex.axis = 0.8)
```



```r
cor_table <- data.frame(
  Descriptor = c("Elevation", "Easting", "Northing"),
  Pearson = sapply(dat[c("Quota", "x", "y")],
                   function(z) cor(z, dat$rain)),
  Spearman = sapply(dat[c("Quota", "x", "y")],
                    function(z) cor(z, dat$rain, method = "spearman"))
)
knitr::kable(cor_table, digits = 3, row.names = FALSE)
```

Descriptor	Pearson	Spearman
Elevation	0.432	0.476
Easting	-0.224	-0.147
Northing	0.019	-0.017

```r
knitr::kable(as.data.frame(table(dat$Nome_Regione)),
             col.names = c("Region", "Stations"))
```

Region	Stations
Basilicata	38
Calabria	149
Puglia	2

Rainfall has a moderate positive association with elevation (Pearson correlation **0.432**). Higher sites tend to receive more rain, but there is considerable scatter. Rainfall decreases somewhat with easting, while its simple linear association with northing is close to zero. This does not mean that northing is irrelevant after accounting for other variables, or that there is no nonlinear spatial pattern.

The regional boxplots are descriptive. Only two stations remain labelled Puglia, so I cannot use that small group to describe rainfall across the whole region. These correlations are associations, not causal effects. Ordinary correlation tests would also need care because observations may be spatially dependent.

2.3 Trend, Gaussianity and the empirical variogram

I compare three simple trend models: elevation alone, coordinates alone, and elevation plus coordinates. For each linear model, the leave-one-out residual is `residual / (1 - leverage)`. This gives its LOOCV RMSE without a long fitting loop. The trends are exploratory; the requested final estimator remains ordinary IDW of rainfall, not regression plus interpolation of residuals.

```r
trend_models <- list(
  Elevation = lm(rain ~ Quota, data = dat),
  Coordinates = lm(rain ~ x + y, data = dat),
  Combined = lm(rain ~ Quota + x + y, data = dat)
)
cutoff <- max(dist(dat[c("x", "y")])) / 2
lag_width <- cutoff / 12
emp_variogram <- function(z) {
  vdat <- data.frame(x = dat$x, y = dat$y, value = z)
  gstat::variogram(value ~ 1, locations = ~ x + y, data = vdat,
                   cutoff = cutoff, width = lag_width)
}
residual_variograms <- lapply(trend_models,
                              function(m) emp_variogram(residuals(m)))
# Same exponential model and starting rule for each residual variogram.
# These are descriptive weighted least-squares fits, not likelihood fits.
```

```r
nugget_fits <- lapply(seq_along(trend_models), function(j) {
  s2 <- var(residuals(trend_models[[j]]))
  gstat::fit.variogram(residual_variograms[[j]],
    gstat::vgm(psill = 0.8 * s2, model = "Exp",
               range = cutoff / 3, nugget = 0.2 * s2), fit.method = 7)
})
stopifnot(!any(vapply(nugget_fits, function(z)
  isTRUE(attr(z, "singular")), logical(1))))
trend_table <- data.frame(
  Trend = names(trend_models),
  CV_RMSE_mm = sapply(trend_models, function(m)
    sqrt(mean((residuals(m) / (1 - hatvalues(m)))^2))),
  Adjusted_R2 = sapply(trend_models, function(m) summary(m)$adj.r.squared),
  Nugget_mm2 = sapply(nugget_fits, function(v) v$psill[v$model == "Nug"])
)
knitr::kable(trend_table, digits = 2, row.names = FALSE)
```

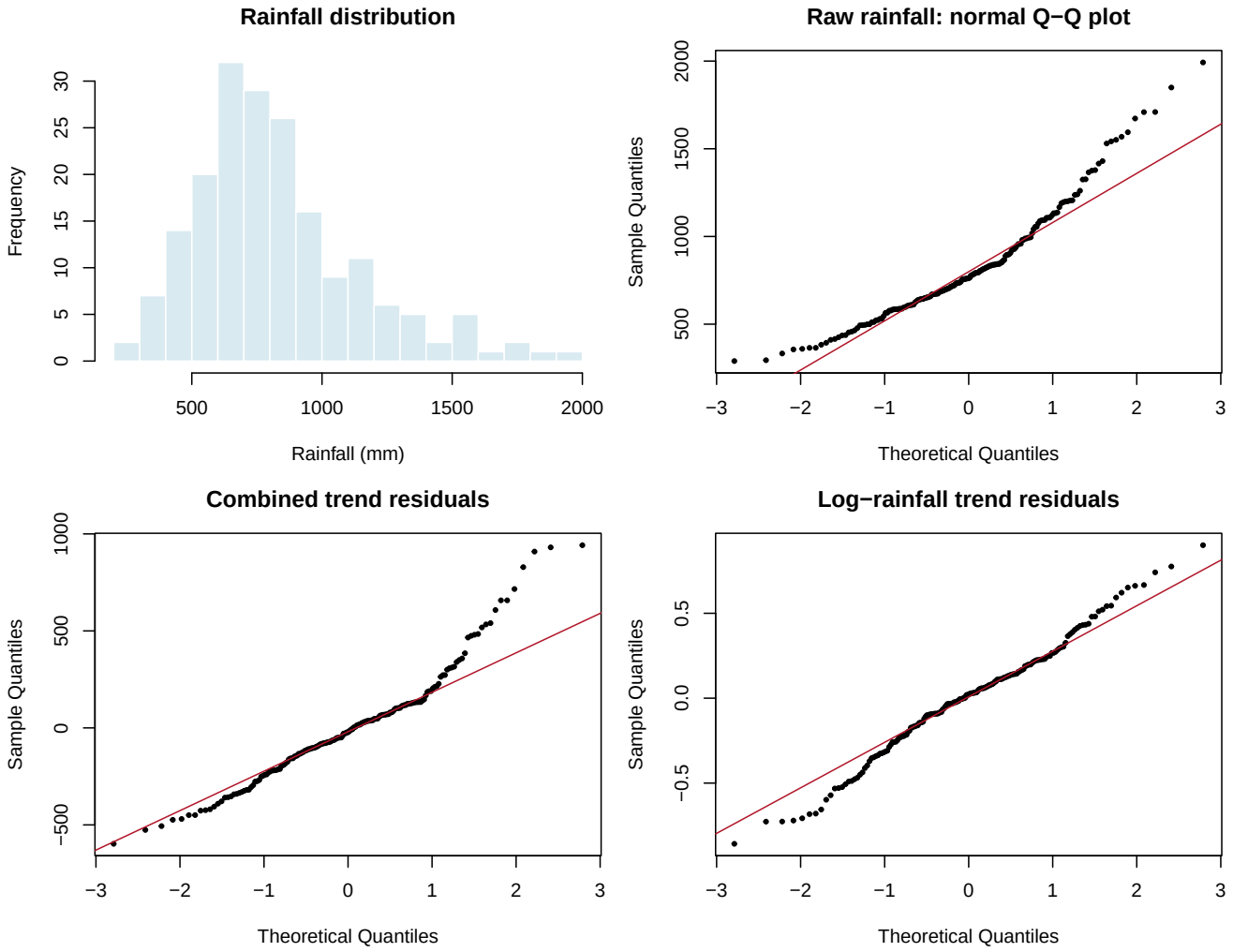
Trend	CV_RMSE_mm	Adjusted_R2	Nugget_mm2
Elevation	288.11	0.18	15325.45
Coordinates	313.64	0.04	19750.03
Combined	280.02	0.24	19882.48

```r
trend <- trend_models$Combined
log_trend <- lm(log(rain) ~ Quota + x + y, data = dat)
```

The combined trend has the lowest CV RMSE among these three candidates (**280.02 mm**), but elevation alone has a lower fitted nugget. I therefore do not choose a trend by minimizing the nugget alone. The combined model is a useful exploratory compromise: it improves the prediction score and includes a simple broad spatial pattern, while leaving substantial unexplained variation. Its adjusted R-squared is only **0.239**.

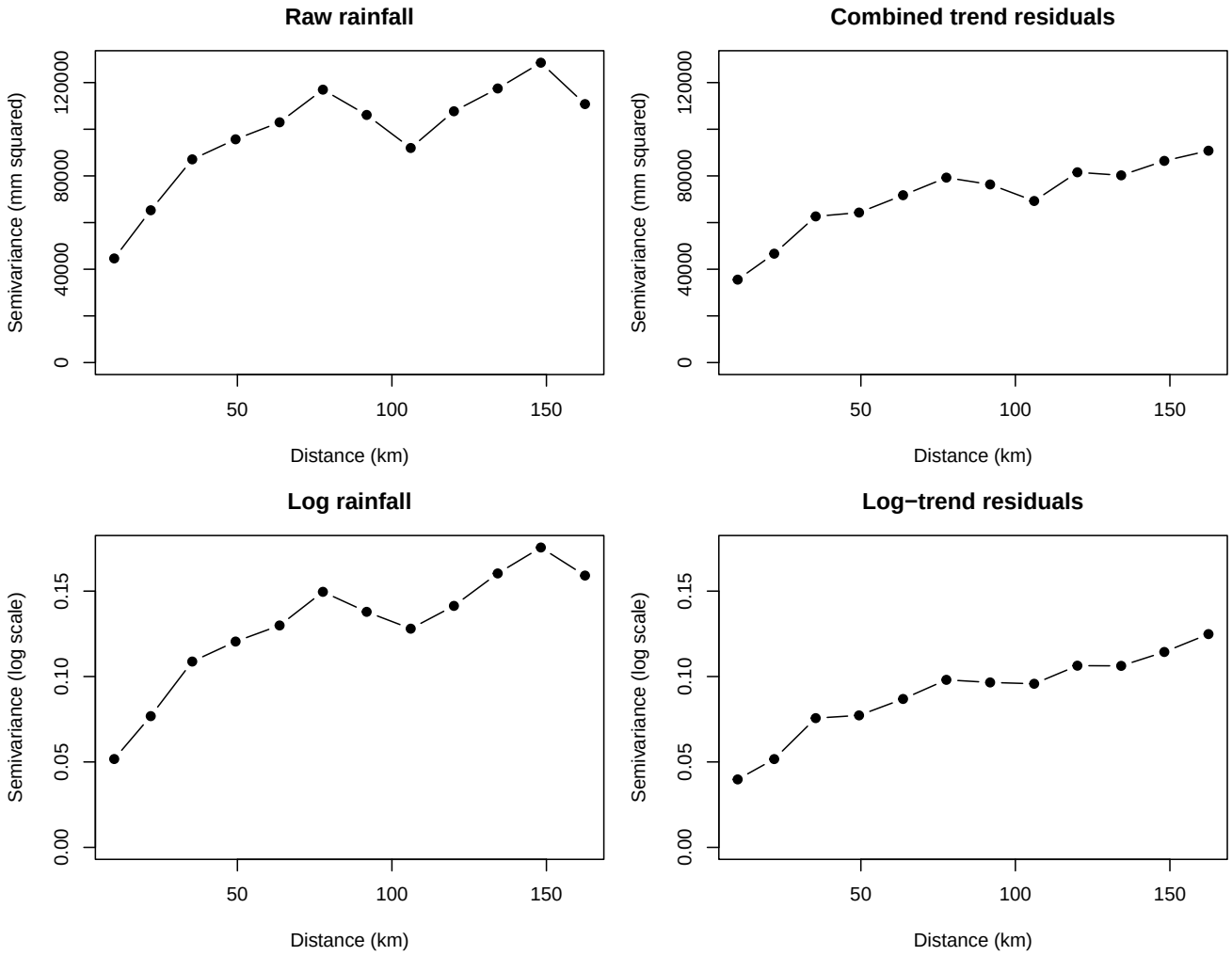
The nugget describes the fitted semivariance as separation approaches zero. It can reflect measurement error and variation below the sampling resolution. Here it is an extrapolation from an empirical variogram, not a directly measured error variance. Its value depends on the covariance family, lag bins and trend. A smaller nugget is not automatically a better model, and the trend comparison itself is exploratory selection rather than independent validation.

```r
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
hist(dat$rain, breaks = "FD", col = "#D9EAF1", border = "white",
     main = "Rainfall distribution", xlab = "Rainfall (mm)")
qqnorm(dat$rain, main = "Raw rainfall: normal Q-Q plot", pch = 19, cex = 0.5)
qqline(dat$rain, col = "#B2182B")
qqnorm(residuals(trend), main = "Combined trend residuals",
       pch = 19, cex = 0.5)
qqline(residuals(trend), col = "#B2182B")
qqnorm(residuals(log_trend), main = "Log-rainfall trend residuals",
       pch = 19, cex = 0.5)
qqline(residuals(log_trend), col = "#B2182B")
```



The raw data and original-scale trend residuals depart from a straight Q-Q line. The log-rainfall residuals look closer to it. A normal-looking marginal distribution would still not establish a Gaussian spatial process. I use the plots as diagnostics rather than treating a normality test as decisive under spatial dependence.

```
v_raw <- emp_variogram(dat$rain)
v_res <- residual_variograms$Combined
v_log <- emp_variogram(log(dat$rain))
v_logres <- emp_variogram(residuals(log_trend))
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
vv <- list(v_raw, v_res, v_log, v_logres)
titles <- c("Raw rainfall", "Combined trend residuals",
            "Log rainfall", "Log-trend residuals")
for (j in seq_along(vv)) {
  same_scale <- if (j <= 2) range(v_raw$gamma, v_res$gamma) else
    range(v_log$gamma, v_logres$gamma)
  plot(vv[[j]]$dist, vv[[j]]$gamma, type = "b", pch = 19,
       ylim = c(0, same_scale[2]), xlab = "Distance (km)",
       ylab = if (j <= 2) "Semivariance (mm squared)" else
         "Semivariance (log scale)", main = titles[j])
}
```




```
knitr::kable(data.frame(Lag = seq_len(nrow(v_raw)),
                        Mean_distance_km = v_raw$dist,
                        Pairs = v_raw$np,
                        Raw_semivariance = v_raw$gamma,
                        Residual_semivariance = v_res$gamma), digits = 2)
```

Lag	Mean_distance_km	Pairs	Raw_semivariance	Residual_semivariance
1	10.13	394	44625.76	35539.29
2	21.95	1019	65276.11	46668.12
3	35.39	1491	87086.77	62644.27
4	49.41	1648	95678.75	64281.95
5	63.65	1611	102970.67	71722.58
6	77.67	1529	116959.35	79264.50
7	91.83	1388	106131.45	76338.85
8	106.08	1273	91966.41	69247.34
9	120.07	1229	107698.45	81547.34
10	134.24	1087	117470.43	80240.54
11	148.19	980	128513.16	86435.85
12	162.49	874	110775.84	90797.14

These are 12 equal-width distance classes up to half the maximum station separation (**169.6 km**). I show the pair counts because a point based on fewer pairs is less stable. Removing the trend changes the variogram, but it does

not remove all variation with distance. The first residual bin has semivariance **35539.3**, compared with **44625.8 mm squared** for the raw observations.

A rising raw variogram may reflect a changing mean as well as spatial dependence. The residual plots help separate these issues, but do not prove stationarity or isotropy. The lower panels repeat the spatial diagnostic on the log scale; their semivariances cannot be compared numerically with the upper panels.

I retain the original response scale for IDW because it estimates rainfall in its original units and does not require Gaussianity or a fitted covariance model. The log analysis above is diagnostic only. If I used log rainfall for prediction, I would need to apply that choice consistently during validation and mapping, and assess the predictions after appropriate back-transformation. Simply exponentiating a predicted log value does not generally estimate the original-scale conditional mean.

3 Choosing the IDW power with cross-validation

IDW predicts a weighted average of the observed rainfall. A station at distance d_i receives weight proportional to d_i^{-p}. A larger power puts more weight on nearby stations. I keep all stations as neighbours and tune only `p`, so the comparison has one changing setting.

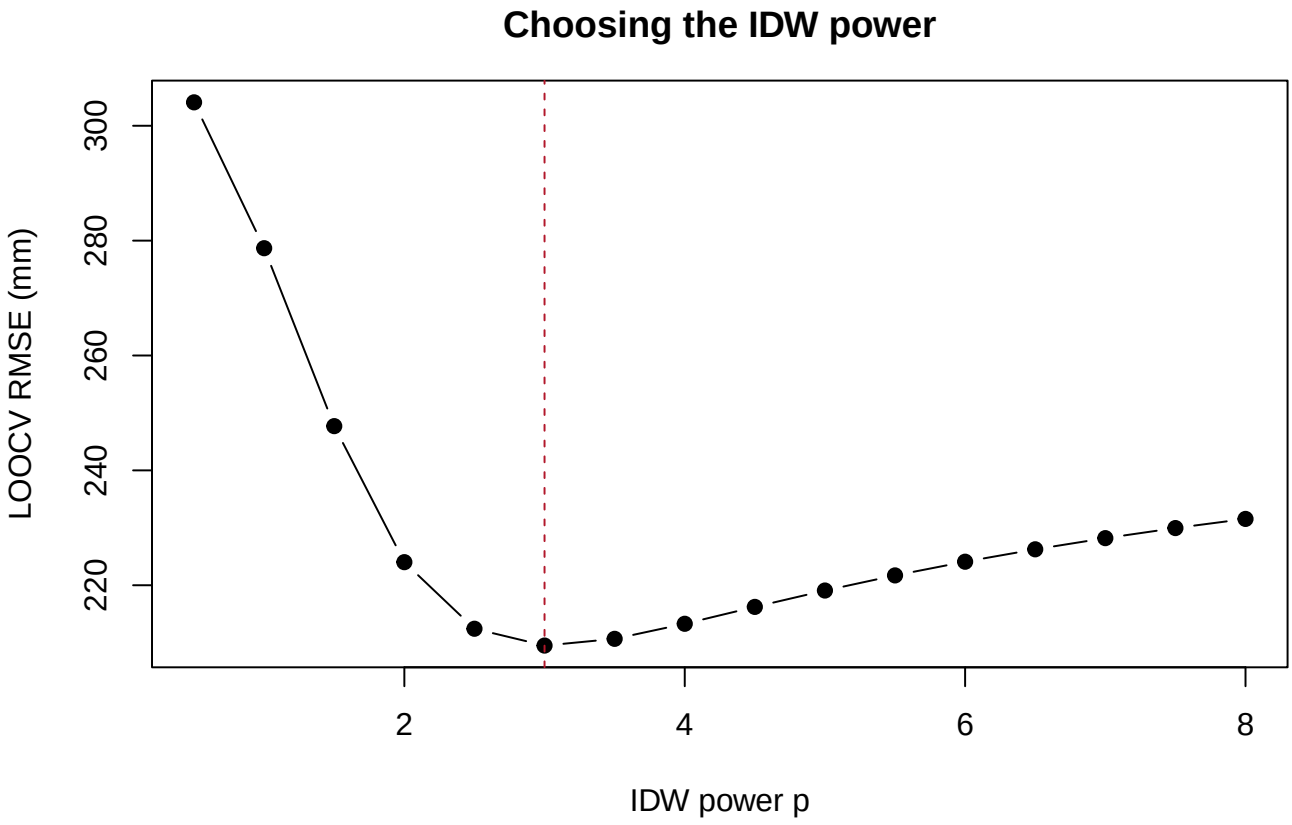
I test powers from 0.5 to 8 in steps of 0.5. Each observation is removed before predicting its value. The chosen power minimizes the leave-one-out RMSE. MAE and mean error provide additional information; mean error is prediction minus observation. The optimum is only the best among the candidates considered.

```
idw_predict <- function(train, newdata, p) {
  gstat::idw(rain ~ 1, locations = ~ x + y, data = train,
             newdata = newdata, idp = p, nmax = Inf,
             debug.level = 0)$var1.pred
}
score <- function(observed, predicted) {
  error <- predicted - observed
  c(RMSE = sqrt(mean(error^2)), MAE = mean(abs(error)), ME = mean(error))
}
choose_p <- function(train, powers = seq(0.5, 8, by = 0.5)) {
  predictions <- matrix(NA_real_, nrow(train), length(powers))
  for (j in seq_along(powers)) {
    for (i in seq_len(nrow(train))) {
      predictions[i, j] <- idw_predict(train[-i, ], train[i, ], powers[j])
    }
  }
  stopifnot(all(is.finite(predictions)))
  scores <- t(apply(predictions, 2, function(z) score(train$rain, z)))
  table <- data.frame(p = powers, scores, row.names = NULL)
  best <- which.min(table$RMSE)
  list(p = powers[best], table = table,
       predictions = predictions[, best])
}
tuning <- choose_p(dat)
best_p <- tuning$p
stopifnot(best_p > min(tuning$table$p), best_p < max(tuning$table$p))
knitr::kable(tuning$table, digits = 2)
```

p	RMSE	MAE	ME
0.5	304.08	234.47	10.43
1.0	278.68	216.08	15.51
1.5	247.70	192.12	16.66
2.0	224.02	172.78	17.63
2.5	212.43	161.30	18.56

p	RMSE	MAE	ME
3.0	209.49	155.90	18.91
3.5	210.67	154.37	18.84
4.0	213.29	154.42	18.62
4.5	216.23	154.77	18.40
5.0	219.08	155.47	18.23
5.5	221.71	156.43	18.10
6.0	224.10	157.45	18.02
6.5	226.25	158.45	17.96
7.0	228.20	159.31	17.92
7.5	229.96	160.06	17.89
8.0	231.55	160.69	17.87

```
par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
plot(tuning$table$p, tuning$table$RMSE, type = "b", pch = 19,
     xlab = "IDW power p", ylab = "LOOCV RMSE (mm)",
     main = "Choosing the IDW power")
abline(v = best_p, lty = 2, col = "#B2182B")
```



The selected power is $\mathbf{p = 3}$, with tuning RMSE **209.49 mm**. For comparison, the common default `p = 2` gives **224.02 mm** under the same validation scheme. Nearby candidate scores should be read from the curve; this is not an exactly known physical parameter.

3.1 A separate check of prediction performance

The minimum above was used to select `p`, so it is not an independent estimate of the selected procedure's accuracy. I add a simple outer five-fold check. For each outer training set, `p` is selected by LOOCV using only that training

set. The untouched outer observations are then predicted. I use a training-mean baseline to check whether IDW improves on a simple non-spatial predictor.

```
set.seed(2026)
fold <- sample(rep(1:5, length.out = nrow(dat)))
outer_prediction <- baseline_prediction <- numeric(nrow(dat))
fold_p <- numeric(5)
for (k in 1:5) {
  test <- fold == k
  train <- dat[!test, ]
  inside <- choose_p(train)
  fold_p[k] <- inside$p
  outer_prediction[test] <- idw_predict(train, dat[test, ], inside$p)
  baseline_prediction[test] <- mean(train$rain)
}
outer_scores <- rbind(
  IDW = score(dat$rain, outer_prediction),
  Training_mean = score(dat$rain, baseline_prediction)
)
knitr::kable(outer_scores, digits = 2)
```

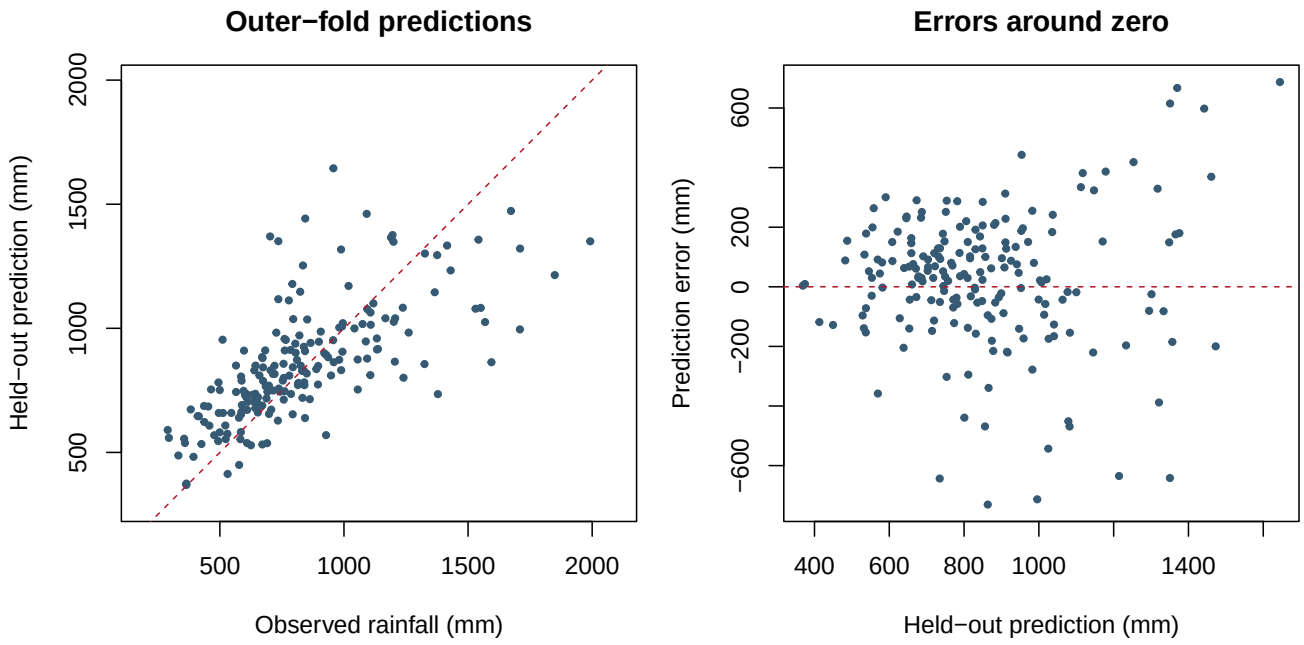
	RMSE	MAE	ME
IDW	228.78	167.53	24.23
Training_mean	320.64	244.30	-0.10

```
knitr::kable(data.frame(Fold = 1:5, Selected_p = fold_p))
```

Fold	Selected_p
1	3.0
2	2.5
3	3.0
4	3.0
5	3.0

The outer IDW RMSE is **228.78 mm**, and MAE is **167.53 mm**. These describe predictions for held-out stations after tuning inside each training fold. They are more useful for discussing predictive performance than the minimum tuning error alone. The training-mean baseline has RMSE **320.64 mm**, so IDW reduces RMSE by about **28.6%** in this comparison. The mean IDW error is **24.23 mm**: predictions are slightly too high on average, even though some wet stations are underestimated. These scores are still based on one random partition, and use fewer training stations per fold than the final map. Differences between fold-specific powers show that the choice can depend on the available stations.

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
lim <- range(dat$rain, outer_prediction)
plot(dat$rain, outer_prediction, xlim = lim, ylim = lim, asp = 1,
     pch = 19, cex = 0.6, col = "#365A73",
     xlab = "Observed rainfall (mm)", ylab = "Held-out prediction (mm)",
     main = "Outer-fold predictions")
abline(0, 1, col = "#B2182B", lty = 2)
plot(outer_prediction, outer_prediction - dat$rain,
     pch = 19, cex = 0.6, col = "#365A73",
     xlab = "Held-out prediction (mm)", ylab = "Prediction error (mm)",
     main = "Errors around zero")
abline(h = 0, col = "#B2182B", lty = 2)
```



IDW averages nearby observations and can smooth local extremes. The prediction plot helps reveal this limitation, which a single RMSE cannot show. Random folds may leave a nearby station in training, so this validation mainly describes prediction among an existing network. It may be optimistic for large unsampled areas. Spatially separated folds and repeated partitions would be useful extensions; I have not presented these as completed analyses.

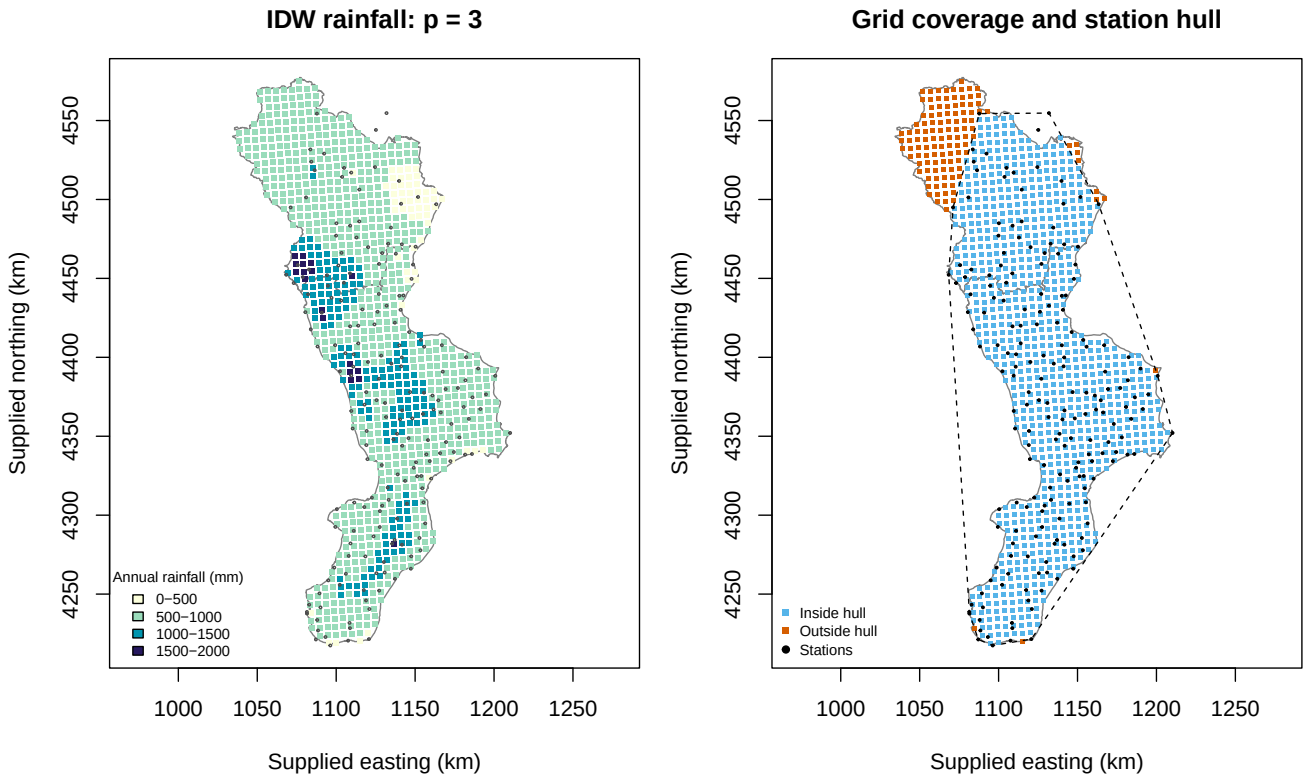
4 Prediction on the supplied land grid

I now use all 189 stations and the power chosen from the full-data LOOCV. I predict at all 1,005 supplied land-grid locations. Their elevations remain in the result, but ordinary IDW uses coordinates and rainfall, not elevation. The EDA therefore identifies an important limitation of this simple estimator.

```
prediction_grid$rain_idw <- idw_predict(dat, prediction_grid, best_p)
h <- chull(dat$x, dat$y)
prediction_grid$inside_hull <- sp::point.in.polygon(
  prediction_grid$x, prediction_grid$y, dat$x[h], dat$y[h]
) > 0
prediction_grid$nearest_km <- vapply(seq_len(nrow(prediction_grid)),
  function(i) min(sqrt((dat$x - prediction_grid$x[i])^2 +
                       (dat$y - prediction_grid$y[i])^2)), numeric(1))
stopifnot(nrow(prediction_grid) == nrow(land_grid),
          all(is.finite(prediction_grid$rain_idw)),
          all(prediction_grid$rain_idw >= min(dat$rain) - 1e-8),
          all(prediction_grid$rain_idw <= max(dat$rain) + 1e-8))
grid_summary <- data.frame(
  Grid_points = nrow(prediction_grid),
  Outside_hull = sum(!prediction_grid$inside_hull),
  Min_prediction_mm = min(prediction_grid$rain_idw),
  Max_prediction_mm = max(prediction_grid$rain_idw),
  Max_nearest_distance_km = max(prediction_grid$nearest_km)
)
knitr::kable(grid_summary, digits = 2)
```

Grid_points	Outside_hull	Min_prediction_mm	Max_prediction_mm	Max_nearest_distance_km
1005	117	291.4	1923.08	46.44

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
map_frame(paste("IDW rainfall: p =", best_p))
points(prediction_grid$x, prediction_grid$y, pch = 15, cex = 0.6,
       col = rain_colour(prediction_grid$rain_idw))
points(dat$x, dat$y, pch = 1, cex = 0.25, col = "grey30")
rain_legend()
map_frame("Grid coverage and station hull")
points(prediction_grid$x, prediction_grid$y, pch = 15, cex = 0.5,
       col = ifelse(prediction_grid$inside_hull, "#56B4E9", "#D55E00"))
polygon(dat$x[h], dat$y[h], border = "black", lty = 2)
points(dat$x, dat$y, pch = 19, cex = 0.25)
legend("bottomleft", c("Inside hull", "Outside hull", "Stations"),
       col = c("#56B4E9", "#D55E00", "black"), pch = c(15, 15, 19),
       bty = "n", cex = 0.7)
```



The rainfall map uses the same colour breaks as the station rainfall map. The squares mark the actual supplied grid locations; I do not fill gaps using a second, unreported interpolation method. Predicted rainfall ranges from **291.4 to 1923.1 mm**.

There are **117 grid points outside the station convex hull**, out of 1005. I retain and flag them. Their estimates extend beyond the sampled spatial envelope and need more care. Even inside the hull, a point may be poorly supported: the hull is not a measure of prediction uncertainty. The largest distance from a grid point to its nearest station is **46.4 km**.

The wettest predicted patches are local rather than a simple increase from south to north. The northern and western edges of the supplied grid include locations beyond the station envelope, where the map is less well supported.

There are no observed rainfall totals at these grid locations, so I cannot calculate an accuracy score for the map itself. The station cross-validation scores provide the available assessment. IDW also gives no model-based prediction intervals; the coverage map should not be interpreted as one.

5 Weak and good AI prompts

5.1 Weak prompt

> Use the rainfall data in R to show that rainfall is spatially dependent. Use IDW with p = 2 and draw a prediction map on the grid.

AI response, recorded for this report:

> I would select one year, plot the station values and an empirical variogram, and use `gstat::idw` with `idp = 2` to predict at the grid coordinates. An increasing variogram can suggest spatial structure, but it does not by itself prove residual spatial dependence: a broad mean trend can also create this pattern. The coordinate system and the choice of grid need checking before interpolation. Using p = 2 gives a conventional starting choice, not a power selected by cross-validation.

My assessment: The response gives a reasonable starting method and avoids claiming that a variogram proves dependence. However, it does not complete the homework: it gives no actual tuning procedure, no data-specific correction and no runnable analysis. The prompt also asks for confirmation rather than an open assessment and fixes a parameter that the assignment asks me to choose. In my executed comparison, p = 2 has LOOCV RMSE 224.02 mm, while the selected p has 209.49 mm. This comparison uses the same observations and folds, but both numbers are tuning results, not separate test results.

5.2 Good prompt

> Using the uploaded RainData.RData and AllGrids.RData, first inspect the objects, years, response, descriptors, coordinate units and available grids. Choose one rainfall year using completeness and station coverage rather than its prediction score; explain the rule. Check missing values and duplicate locations. Locate the Puglia-labelled station that belongs to Calabria using the supplied spatial information, correct its label and retain its records. Draw a station map and examine annual rainfall against elevation, location and region. Examine the distribution, simple trends and raw/residual variograms with explicit distance classes and pair counts. Explain what these checks can and cannot establish; consider predictive score and nugget together when discussing trends. Do not require normality just to use IDW. Write simple runnable R code using gstat::idw to choose p by cross-validation, stating the candidate powers and error score. Keep tuning separate from any reported assessment of the selected procedure. Predict at the supplied grid that actually matches the rainfall data, explain any mismatch between files, and flag locations with weak spatial support. Give only calculated numerical results, use cautious conclusions and include the code and figures in an R Markdown report that compiles to PDF.

AI response, recorded for this report:

> The rainfall file contains observations from 1970–1982 and a matching land grid. The other file contains marine predictors for 1995–2010 and should not be silently joined to these rainfall observations. I will use the existing land grid in RainData.RData. The earliest year tied for the greatest number of complete records is 1970, with 189 stations. Station 4367 is the Puglia-labelled station inside the supplied Calabria outline; its label can be corrected without changing its measurement.
>
> Elevation is positively associated with rainfall, but a simple trend leaves substantial variation. A raw variogram can combine mean variation and residual dependence, so I will compare it with residual variograms and show pair counts. These plots do not prove stationarity. The residual nugget is model-dependent, and normal-looking residuals would not prove joint Gaussianity. IDW itself does not require normality, so I will keep the final predictions on the original rainfall scale and label the log analysis as a diagnostic.
>
> I will implement `idw_predict` and `choose_p` using gstat, leave each station out before predicting it, and choose the candidate power with minimum RMSE. An outer five-fold check will repeat tuning inside each training fold so that its held-out observations do not select p. I will then use all 1970 observations

for the final land-grid predictions and flag points outside the station hull. Random-fold validation may be optimistic for distant unsampled areas, and there is no grid accuracy score without observed grid rainfall. The runnable implementation and calculated results are in the report above.

My assessment: This response addresses the actual files rather than assuming that every uploaded grid is suitable. It distinguishes trend from residual dependence, explains the limits of the variogram, and avoids imposing a Gaussian model on IDW. It also connects the tuning method to an explicit score and keeps outer test observations out of parameter selection. These choices make the analysis easier to reproduce and check. It is still conditional on the assumed measurement units and common projected coordinates; the files do not supply complete metadata. More words alone would not solve those limitations.

The responses above were generated by ChatGPT/Codex during preparation of this report, in the same conversation and with access to the data inspection. They are newly recorded responses, not an independent or blinded prompt experiment. The weak response is not deliberately made statistically incorrect. The code above is the implemented good-response workflow; the fixed-p comparison was calculated within that code, rather than presented as a separate historical AI run.

6 Final discussion and reproducibility

For this year, elevation helps describe rainfall differences, while a simple trend does not explain everything. Cross-validation gives a transparent choice of IDW power, and the final map uses the existing rainfall grid. I would treat predictions near sparse areas and outside the station hull with particular care. The selected year, ordinary IDW's omission of elevation, one random outer partition, and incomplete metadata limit how widely I can generalize the result.

AI assistance and responsibility. AI assisted with the prompts, statistical workflow, R code, explanations and report preparation. The code was executed in the working R environment to produce the reported results. The student-style discussion is also an AI-assisted draft. I remain responsible for reviewing the assumptions, checking the code and understanding the conclusions before submission. This statement does not claim student edits or checks that have not yet been made for Homework 2.

The functions used here are documented in `help("idw", package = "gstat")`, `help("variogram", package = "gstat")`, `help("fit.variogram", package = "gstat")`, and `help("point.in.polygon", package = "sp")`. In particular, the variogram fits here use weighted least squares, and IDW is deterministic; neither is Bayesian analysis. No Bayesian or likelihood-based spatial model is fitted in this task.

```
versions <- data.frame(
  Component = c("R", "gstat", "sp", "knitr", "rmarkdown"),
  Version = c(as.character(getRversion()),
    vapply(c("gstat", "sp", "knitr", "rmarkdown"),
           function(p) as.character(packageVersion(p)), character(1)))
)
knitr::kable(versions, row.names = FALSE)
```

Component	Version
R	4.5.0
gstat	2.1.6
sp	2.2.0
knitr	1.50
rmarkdown	2.29